

- Q.29 Describe how maintaining the work area safe and secure during PV installation. (CO4)
- Q.30 Enlist component used in a grid photovoltaic power generation system. (CO1)
- Q.31 What is the importance of better communication between management and individual at workplace of PV installation? (CO3)
- Q.32 Explain the importance of individuals role on the PV industry overflow. (CO3)
- Q.33 Write five uses of Solar panel. (CO1)
- Q.34 How many types of semiconductors or solar cells used to fabricate PV cells? (CO2)
- Q.35 Write short note on maintenance procedure of solar panel. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain different types of solar panel mounting structure and their merits and demerits. (CO1)
- Q.37 Describe series and parallel connection of solar cell and also define angle of tilt of solar panel. (CO2)
- Q.38 Explain advantages and disadvantages of solar energy system over conventional power system. (CO4)

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Roll No.

5th Sem / Elect

Sub : Solar Panel Installation & Maintenance

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Fall protection system can include which these of following. (CO4)
- Personal fall arrest system
 - Guardrail system
 - Safety nets
 - All of above
- Q.2 Death due to electrical shock is called (CO4)
- Electrocution
 - Heart attack
 - Brain failure
 - None of the above
- Q.3 Solar tracking system is classified its (CO5)
- Degree of rotation
 - Axis for rotation
 - Both of these
 - None of these
- Q.4 The single solar cell voltage is about (CO1)
- 0.2 v
 - 0.5v
 - 1.0v
 - 2.0v
- Q.5 Components of PV solar system (CO2)
- PV array
 - Batteries and charge controller
 - Inverter
 - All A, B & C

- Q.6 Most widely used solar material is (CO1)
 a) Arsenic b) Silicon
 c) Cadmium d) Steel
- Q.7 When installing a solar module on a roof, things to be considered are (CO4)
 a) Obstructions on roof
 b) The condition of the roof
 c) Far shading
 d) All of the above
- Q.8 Energy in the form of the heat and light is obtained by (CO2)
 a) Biomass b) Sun
 c) Fossil fuels d) Wind
- Q.9 To make sure all PV projects are completed safely, employers should have policies and procedures (CO3)
 a) Provide necessary equipment to do job right
 b) Provide training to all workers
 c) Allow workers to work fast and profitably
 d) All of the above
- Q.10 Which cell is used to convert solar energy directly into electrical energy (CO9)
 a) Dry cell b) Battery
 c) Photovoltaic cell d) None of these

SECTION-B

- Note:** Objective type questions. All questions are compulsory. (10x1=10)
- Q.11 Name the material used for making solar panel mounting structure. (CO4)

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- Q.12 Write one benefit of individual role in workflow. (CO3)
- Q.13 Write one function of personal management. (CO3)
- Q.14 Give the two examples of monetary rewards. (CO3)
- Q.15 Define solar energy. (CO1)
- Q.16 Write one cause of fire accident at workplace of PV installation. (CO4)
- Q.17 The function of charge controller is _____. (CO2)
- Q.18 The efficiency of solar panel is about _____. (CO1)
- Q.19 Write one point to handle the Hazardous material during PV installation. (CO4)
- Q.20 Define Tilt angle. (CO2)

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 What is solar microgrid system and what are their major components? (CO4)
- Q.22 Write a short note on sunlight assessment. (CO2)
- Q.23 Explain the benefits of team coordination in PV installation. (CO3)
- Q.24 What are the safety precautions used to avoid fire accidents at workplace? (CO4)
- Q.25 Explain the concept of solar tracking system. (CO5)
- Q.26 Explain any five instructions regarding working safety with solar PV installation. (CO1)
- Q.27 What are the significance of voltage, amperes and watts in solar cell construction? (CO2)
- Q.28 Explain the parallel connection of PV panel. (CO2)

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